

FINALTERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Ref No:

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

Scatterplots are used only for

- ▶ quantitative variables
- ▶ **multivariables**
- ▶ surplus variable
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason of the error?

- ▶ Array1 and Array2 have different number of data points.
- ▶ **Either Array1 or array2 is empty.**
- ▶ Array or reference argument contains text, logical values or empty cells.
- ▶ The arguments are names, arrays, or references that contain numbers.

Question No: 3 (Marks: 1) - Please choose one

_____ A
college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

- ▶ 1260
- ▶ **210**
- ▶ $10C6 * 6!$
- ▶ $10C5 * 6$

Question No: 4 (Marks: 1) - Please choose one

_____ An
insurance company wants to predict sales from the amount of money they spend on advertising. Which would be the independent variable?

- ▶ (i) sales
- ▶ **(ii) advertising**

- ▶ (iii) insufficient information to decide
- ▶ (iv) Both (i) and (ii) are correct.

Question No: 5 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- ▶ bar graph
- ▶ pie graph
- ▶ pictograph
- ▶ line graph

Question No: 6 (Marks: 1) - Please choose one

In a positively skewed distribution

- ▶ The mean, median, and mode are all equal.
- ▶ The mean is larger than the median
- ▶ The median is larger than the mean.
- ▶ The standard deviation must be larger than the mean or the median.

Question No: 7 (Marks: 1) - Please choose one

Which one of the following is not a component of the multiplicative time series model?

- ▶ trend
- ▶ irregular variation
- ▶ regression trend
- ▶ seasonality
- ▶ cyclicity

Question No: 8 (Marks: 1) - Please choose one

How many number of times will the digit '7' be written when listing the integers from 1 to 1000?

- ▶ 271
- ▶ 300
- ▶ 252
- ▶ 304

Question No: 9 (Marks: 1) - Please choose one

How many arrangements can be made of the letter BUSINESS

- ▶ 6723
- ▶ 6725
- ▶ 6720
- ▶ 6721

Question No: 10 (Marks: 1) - Please choose one

The moving averages represent -----

- ▶ Time series variations
- ▶ Co-efficient of variations
- ▶ Statistical Dispersion
- ▶ Absolute deviation

Question No: 11 (Marks: 1) - Please choose one

_____ The class frequency is

- ▶ The number of observations in each class
- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

Question No: 12 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the standard deviation is approximately

- ▶ 4.75
- ▶ 4.55
- ▶ 3.03
- ▶ 3.75

Question No: 13 (Marks: 1) - Please choose one

_____ The experimental region is the range of the previously observed values of the dependent variable.

- ▶ False
- ▶ True

Question No: 14 (Marks: 1) - Please choose one

Which of the following is not a violation of the independence assumption?

- ▶ Negative autocorrelation
- ▶ A pattern of cyclical error terms over time
- ▶ Positive autocorrelation
- ▶ A pattern of alternating error terms overtime
- ▶ A random pattern of error terms over time

Question No: 15 (Marks: 1) - Please choose one

_____ All of the following are assumptions of the error terms in the simple linear regression model except

- ▶ normality.
- ▶ error terms with a mean of zero.
- ▶ constant variance.
- ▶ variance of one.

Question No: 16 (Marks: 1) - Please choose one

_____ If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- ▶ slope, intercept
- ▶ intercept, slope
- ▶ slope, regression coefficient
- ▶ radius, intercept

Question No: 17 (Marks: 1) - Please choose one

_____ If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ Midterm has greater variation in marks than Final term
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above.

Question No: 18 (Marks: 1) - Please choose one

_____ If A and B are two mutually exclusive events, then

- ▶ $P(A \cup B) = P(A).P(B)$
- ▶ $P(A \cap B) = P(A) + P(B)$

► $P(A \cup B) = P(A) + P(B)$

► $P(A \cup B \cup C) = P(A) + P(B)$

Question No: 19 (Marks: 1) - Please choose one

_____ The

$$\rho_{x,y} = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y}$$

equation for the correlation coefficient is

where

► $0 \leq \rho_{x,y} \leq 1$

► $-1 \leq \rho_{x,y} \leq 1$

► $1 \leq \rho_{x,y} \leq 0$

► $-1 \leq \rho_{x,y} \leq 0$

Question No: 20 (Marks: 1) - Please choose one

After the merchant buys merchandise, it is sold at a higher price called the _____

► Sale price

► Revenue discount

► Selling price

► Cost price

Question No: 21 (Marks: 1) - Please choose one

_____ A
reduction of the amount due on an invoice is called a _____ .

► Trade discount

► Net discount

► Cash discount

► Unearned discount

Question No: 22 (Marks: 1) - Please choose one

_____ If
the basic salary of an employee is 13000 what is the amount of allowances he is getting for his conveyance?

- ▶ 325
- ▶ 260
- ▶ 765
- ▶ 500

Question No: 23 (Marks: 1) - Please choose one

_____ The minimum value of the correlation coefficient r can be

- ▶ -infinity
- ▶ 0
- ▶ -1
- ▶ 1

Question No: 24 (Marks: 1) - Please choose one

Total Provident Fund added to the employee's fund is ----- of the basic salary.

- ▶ 1 / 11 th
- ▶ 2 / 11 th
- ▶ 9.09 %
- ▶ 9.99%

Question No: 25 (Marks: 1) - Please choose one

If I is an identity matrix then it must also be a

- ▶ rectangular matrix
- ▶ row matrix
- ▶ column matrix
- ▶ scalar matrix

Question No: 26 (Marks: 1) - Please choose one

_____ If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ▶ None of these

Question No: 27 (Marks: 1) - Please choose one

_____ If
 A is the matrix of dimension 2×3 and I is the identity matrix of dimension 3×3 . Then which of the following is true

► $AI = I$

► $AI = A$

► $A + I = A$

► $A + I = I$

Question No: 28 (Marks: 1) - Please choose one

_____ If
the cost & selling price of a pen are Rs.12 & Rs.15 respectively, profit percentage is:

► 33.33%

► 25%

► 20%

► 10%

Question No: 29 (Marks: 1) - Please choose one

_____ The net price of a computer table is 2500 and list price 3000. The trade discount will be

► 13%

► 17%

► 21%

► 25%

Question No: 30 (Marks: 1) - Please choose one

_____ Ogives at different values determine

► limits .

► frequencies.

► cumulative frequencies.

► both (2) & (3).

Question No: 31 (Marks: 1) - Please choose one

While using Frequency function ,one always selects

► one cell more than data array.

► one cell more than bins array .

► at most 20 cells.

► random number of cells.

Question No: 32 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

► $m+n$

► $m-n$

► $m*n$

► n/m

Question No: 33 (Marks: 1) - Please choose one

The variance of the 3,3,3,3,3 is

► Zero .

► One.

► Twice the standard standard deviation.

► Half of the range.

Question No: 34 (Marks: 1) - Please choose one

Which of the following is most probably is an example of impossible event while tossing of a coin

► Coming of head

► Coming of tail

► Coming of erected coin

► None of these

Question No: 35 (Marks: 1) - Please choose one

For

two tail test, when $\alpha = 0.10$ the value of Z is

- ± 1.96
- ± 1.645
- ± 2.326
- ± 2.575

Question No: 36 (Marks: 1) - Please choose one

For

the set of data 1, 2, 3, 4, 5, 2, 1, 6, 8, the mode is given by

- 1
- 2
- 1 and 2
- 3

Question No: 37 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

- True
- False

Question No: 38 (Marks: 1) - Please choose one

Linear trend is calculated as $Y_t = 30.5 + 0.55t$. The trend projection for period 15 is

- 11.25
- 28.50
- 38.75
- 44.25

Question No: 39 (Marks: 1) - Please choose one

.....is a deduction from the list price of goods provided by a business in return for payment within a specified time.

- trade discount
- cash discount
- credit discount

Question No: 40 (Marks: 1) - Please choose one

Ogive of a statistical data can be drawn by

► a) using the cumulative frequency of the distribution

► b) frequency of the distribution

► c) both (a) & (b)

► d) None of these.

Question No: 41 (Marks: 2)

Find harmonic mean (HM) of 10,12,14,17.

Solution:

X	1/X
10	0.1
12	0.08333
14	0.07142
17	0.05882
	$\sum \frac{1}{x} = 0.031357$

$$\text{Harmonic Mean} = \frac{n}{\sum \frac{1}{x}}$$

$$\text{Harmonic Mean} = \frac{4}{0.031357}$$

$$\text{Harmonic Mean} = 12.756$$

Question No: 42 (Marks: 2)

If you toss a die and observe the number of dots that appears on top face then write the events that the even number occurs.

Solution:

Number of Possible outcomes = 6

Number of Events = 3

P= Number of events/Number of all Possible outcomes.

P= 3/6

Question No: 43 (Marks: 2)

Define Cumulative Poisson distribution.

Solution:

A cumulative Poisson distribution is used to calculate the probability of getting atleast n successes in a Poisson experiment. Here, n is the Poisson random variable which refers to the number of success.

Formula:

$$P(x \leq n) = P(x = 0) + P(x = 1) + \dots + P(x = n)$$

where, $P(x = 0)$ and $P(x = 1)$ is calculated using Poisson distribution formula.

Question No: 44 (Marks: 3)

Four friends take an IQ test. Their scores are 96, 100, 106, 114. Which of the following statements is true? Give reason.

I. The mean is 103.

II. The mean is 104.

III. The median is 100.

IV. The median is 106.

(A) I only

(B) II only

(C) III only

(D) IV only

(E) None is true

Solution:

The correct answer is (B). The mean score is computed from the equation:

$$\text{Mean score} = \Sigma x / n = (96 + 100 + 106 + 114) / 4 = 104$$

Since there are an even number of scores (4 scores), the median is the average of the two middle scores. Thus, the median is $(100 + 106) / 2 = 103$

Question No: 45 (Marks: 3)

What is the probability that a bag filled by the machine weighs less than 515 g?

$$z = (515 - 510)/2.5 = 2.0$$

(Use the table given below)

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936

Solution:

Method is available in Lecture 40.....

$$=0.5+0.4772 = 0.9772$$

Question No: 46 (Marks: 3)

State three important properties of regression line

Solution:

There are three important properties of regression line are as follows.\

1. Slope
2. Intercept
3. Straight Line

Slope:

The slope is the vertical distance divided by the horizontal distance between any two points on the line, which is the rate of change along the regression line.

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{(n \sum x^2 - (\sum x)^2)}$$

Intercept:

Use the INTERCEPT function when you want to determine the value of the dependent variable when the independent variable is 0 (zero).

Straight Line:

The simplest is the Simple Linear. Regression Model or a relationship between variables that can be represented by a straight line equation.

$$Y = a + bX$$

Question No: 47 (Marks: 5)

Calculate the mean, median and mode for the following set of data 5, 6, -5, 8, 19, 2, 1, 6, 25

Solution:

X

1

2

5

-5

6

6

8

19

25

67

Mean

$$\bar{X} = \frac{\sum x}{n}$$

$$\bar{X} = \frac{67}{9}$$

$$\bar{X} = 7.44$$

Median

$$\tilde{X} = \frac{(n+1)}{2}$$

$$\tilde{X} = \frac{(9+1)}{2}$$

$$\tilde{X} = \frac{(10)}{2}$$

$$\tilde{X} = 5$$

Mode

$$\hat{X} = \text{Most frequent Value}$$

$$\tilde{X} = 6$$

Question No: 48 (Marks: 5)

Solve the equation: $C_2^5 + 3x = -C_3^5$

Solution:

$$C_2^5 + 3x = -C_3^5$$

$$3x = -C_3^5 - C_2^5$$

$$3x = -\frac{5!}{3!(5-3)!} - \frac{5!}{2!(5-2)!}$$

$$3x = -\frac{120}{6(2)!} - \frac{120}{2(3)!}$$

$$3x = -\frac{120}{6(2)} - \frac{120}{2(6)}$$

$$3x = -\frac{120}{12} - \frac{120}{12}$$

$$3x = -10 - 10$$

$$3x = -20$$

$$x = -\frac{20}{3}$$

$$x = -6.67$$

Question No: 49 (Marks: 5)

The ages of 5 students in a population are 12, 16, 10, 14 and 13. Considering all possible samples of size two which can be drawn with replacement from this population, find

- The mean age.
- The standard deviation of the students.

Solution:

X	$X - \bar{X}$	$(X - \bar{X})^2$
10	-3	9
12	-1	1
13	0	0
14	1	1
16	3	9
$\sum x = 65$		$\sum (X - \bar{X})^2 = 20$

$$\bar{X} = \frac{\sum x}{n}$$

$$\bar{X} = \frac{65}{5}$$

$$\bar{X} = 13$$

$$S.D = \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$$

$$S.D = \sqrt{\frac{20}{5}}$$

$$S.D = \sqrt{4}$$

$$S.D = 2$$

Question No: 50 (Marks: 10)

What percentage of bags weighs between 512 and 515?

Where mean and standard deviation are 510 and 2.5 respectively.

(Use the following chart:

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936

Solution:

This Question Complete solution given in Lecture 40 Last Example....